
Final Report Bayonetta

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Abstract

This study investigates how temporary environmental disruptions affect predator–prey population dynamics through an agent-based simulation of a simplified ecosystem. Using the Violet Simulator, we modeled interactions between grass (resource), rabbits (prey), and foxes (predators), integrating energy consumption, reproduction, and death mechanisms. A key component of the simulation involved a time-limited ecological crisis in which grass ceased to regrow, simulating events such as droughts or wildfires. We compared population-level outcomes between baseline and crisis conditions across ten simulation runs. Results revealed that temporary resource depletion significantly disrupted the balance between the predator and the prey, leading to a paradoxical increase in prey populations and a sharp decline in predator numbers. These findings suggest that short-term environmental shocks can have unequal effects on different species roles within the food chain, reshaping community structure even in systems that ultimately recover. Our work highlights the importance of individual behavior and local interactions in understanding ecosystem resilience and offers a foundation for extending agent-based models to more complex ecological and environmental scenarios.

1 Introduction

Understanding predator–prey interactions is important for studying the stability and resilience of ecosystems. In natural food chains, even small changes in the environment can cause major effects across different species. This project uses an agent-based simulation to explore how a short-term natural disaster affects the balance between predators and prey.

We used the Violet Simulator, a Python-based tool for modeling group behavior, to create a simple ecosystem made up of rabbits (prey), foxes (predators), and grass (the food source for rabbits). Rabbits eat grass to stay alive and reproduce, while foxes hunt rabbits to survive. All animals lose energy over time and can die from starvation or old age.

A key feature of our study is the introduction of a temporary environmental crisis. During this event, all grass is removed from the environment and cannot grow back for a short time. This simulates real-world disasters like droughts or wildfires and helps us understand how such disruptions affect survival and recovery.

The research question for this project is: How does a temporary environmental crisis affect predator–prey population dynamics, extinction risks, and recovery?

To answer this, we built on earlier research that uses both mathematical and agent-based models to study population interactions (Wilensky and Rand, 2018; Vasconcelos et al., 2020). Other recent studies show that environmental changes and resource loss can strongly affect whether species survive or go extinct (Chen et al., 2019; Venturino, 2014). By including a timed disaster in the simulation, we were able to test how the system responds under stress.

We compared the simulation results with and without the crisis. We collected time-series data for rabbit and fox populations and analyzed how the disaster affected their behavior. The results show clear differences in how stable the populations were, how likely they were to recover, and how much their numbers changed depending on whether the grass was removed.

2 Methodology

Overview of the Agent-Based Modeling Approach

This research utilizes an agent-based modeling (ABM) approach to simulate predator-prey dynamics based on the classic Lotka-Volterra equations. Unlike traditional mathematical methods, ABM allows us to explore detailed individual behaviors and spatial interactions, capturing emergent phenomena resulting from different-level interactions.

Agent Types and Characteristics

Our model includes three distinct agent types:

1. **Foxes (Predators):** Fox agents have an energy attribute, which decreases over time. They gain energy by consuming rabbits. A fox reproduces asexually with a specific probability when it successfully captures and eats a rabbit. Foxes can also die spontaneously based on their energy depletion or a probabilistic death event.
2. **Rabbits (Prey):** Rabbit agents have energy attributes that also decrease over time. Rabbits replenish energy by consuming grass agents, which regrows periodically. Rabbits reproduce asexually with a certain probability, independent of direct predator interaction. Rabbits are only removed from the simulation if consumed by foxes.
3. **Grass (Resource):** Grass is implemented as a static agent, distributed spatially across the environment. Grass regrows after a fixed time interval once consumed, providing a renewable resource essential to the rabbit population's survival.

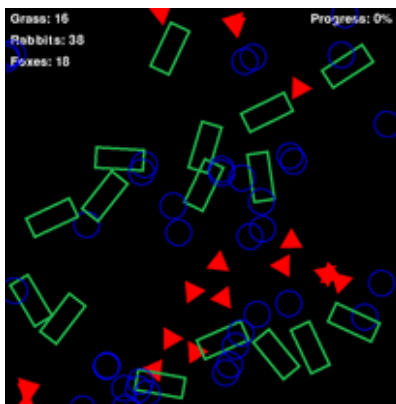


Figure 1: Visual representation of agents in the simulation environment. Foxes are shown as red triangles, rabbits as blue circles, and grass as green rectangles.

Agent Interactions and Behaviors

Agents interact based on spatial proximity, governed by simple decision rules:

- Predation: When a fox encounters a rabbit within a specified proximity, the rabbit is removed from the simulation ("eaten"), and the fox's energy level increases.
- Grazing: Rabbits replenish their energy by consuming grass when encountered within proximity. Consumed grass enters a regrowth timer before becoming available again.

Energy Dynamics and Survival

Energy dynamics introduce additional realism to the model:

- Each agent (fox and rabbit) has an energy value that decreases incrementally at each simulation step.
- Foxes gain energy exclusively by consuming rabbits; rabbits gain energy by consuming grass.
- Agents die either spontaneously based on predefined probabilities or when their energy reaches zero.

Crisis Scenario Implementation

An additional complexity introduced into the model is the "grass crisis scenario":

- At randomly selected simulation intervals, grass growth halts entirely for a specified period, simulating environmental stress.
- During these intervals, rabbits face severe energy depletion risks, indirectly impacting the fox population through resource scarcity.

Metrics for Analysis

To analyze our results, we use the following metrics:

- Population Counts: The number of foxes, rabbits, and grass agents per simulation step, providing insight into population dynamics over time.
- Phase-Space Analysis: Fox versus rabbit population plots reveal cyclical dynamics and equilibrium states.
- Energy Distribution: Periodic snapshots of agent energy distributions to assess the impact of energy dynamics on population stability.

Visualization and Data Processing

- Population data from multiple runs was aggregated and averaged to create representative visualizations, including mean line graphs, phase-space diagrams, and violin plots to illustrate variability and distribution of population dynamics.
- Python libraries such as Pandas, NumPy, and Matplotlib were utilized for data analysis and visualization.

3 Experiments

We conducted ten simulation runs using the Violet Simulator to model predator–prey dynamics under two environmental conditions: with and without a temporary grass-regrowth crisis. Each run consisted of an ecosystem populated by foxes (predators), rabbits (prey), and grass (the

rabbits' food source). Simple local rules for energy use, reproduction, and death ruled certain agents.

In the disaster condition (five runs), grass regrowth was completely disabled during a randomly chosen time window, simulating an acute environmental shock such as a drought or wildfire. In the control condition (five runs), the environment remained stable throughout. Each simulation ran for 25,000 time steps.

Agent population counts (foxes, rabbits, grass) were recorded at each step. We then computed per-run summary metrics such as mean population size, minimum values, and total area under the population curve (AUC). These values formed the basis for statistical comparison between the two conditions.

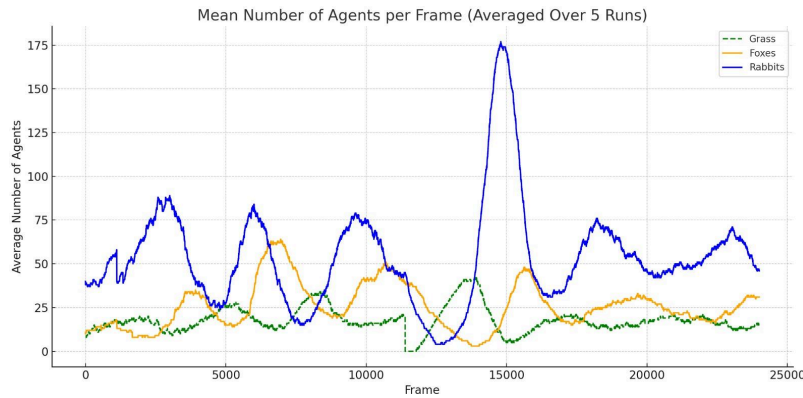


Figure 2: The mean number of agents per frame for the first 5 runs (disaster runs)

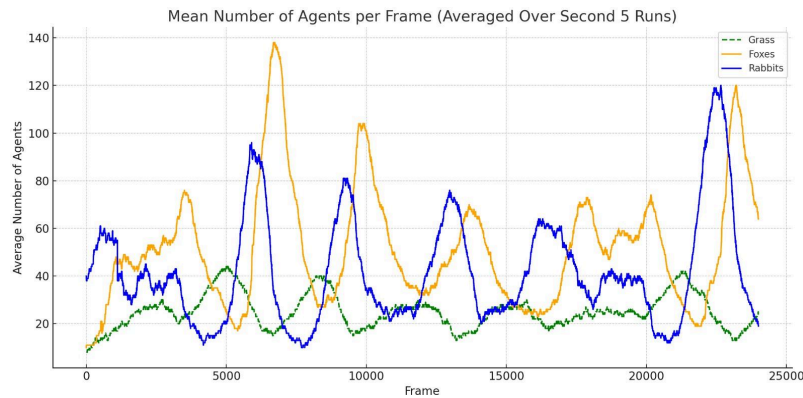


Figure 3: The mean number of agents per frame for the second 5 runs (control runs)

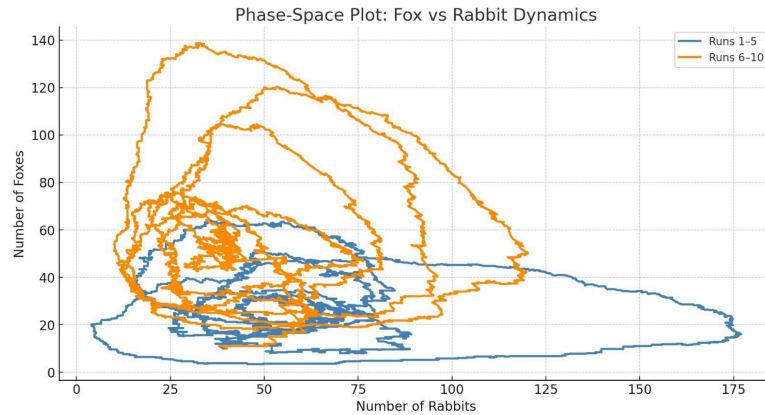


Figure 4: Another way of representing agent populations with both disaster (blue) and control (orange) runs

3.1 Results

We analysed three key response variables: mean population size, cumulative abundance, and extinction frequency across ten simulations (five disaster, five control). Unequal variances made Welch's two-sample t-test appropriate for the continuous metrics.

3.1.1 Mean Population Size

Species	Disaster (M +/- SD)	Control (M +/- SD)	Welch t (df)	p
Rabbits	54.3 +/- 3.5	42.0 +/- 3.3	7.76 (5.6)	< 0.001
Foxes	26.1 +/- 9.0	51.8 +/- 4.2	-6.37 (4.6)	0.001

The average number of rabbits in the disaster condition was 54.3, with a standard deviation of 3.5 (values across the 5 runs were close to this average). The average number of rabbits in the control condition was 42.0, with slightly less variation. Since $p < 0.05$, this result is statistically significant.

The average number of foxes in the disaster condition was 26.1, with a higher standard deviation of 9.0, indicating more variation across the 5 runs. In the control condition, foxes averaged 51.8 with lower variation between runs. The Welch's t-statistic is -6.37 with 4.6 degrees of freedom. A negative value shows the average in the crisis group is lower than in the control. Since $p < 0.05$, the result is statistically significant.

Rabbit populations were significantly larger during the grass-crisis runs, whereas fox populations were significantly smaller.

3.1.2 Cumulative Abundance (Area Under the Curve)

Species	Disaster (M)	Control (M)	Welch t (df)	p
Rabbits	40 666	32 004	7.84 (5.6)	< 0.001
Foxes	18 890	36 236	-6.29 (4.9)	0.001

The total rabbit abundance over time was 40,666 in the disaster condition. This reflects both the size of the population and how long it persisted at those levels. In the control condition, rabbits had a lower total abundance over time. The t-value is high and positive, indicating a large

difference in rabbit AUC between conditions. This very low p-value means the difference is statistically significant.

Foxes had a lower cumulative abundance during the crisis. In the control runs, foxes maintained a much higher total population over time. The negative t-value shows that the disaster condition had significantly lower AUC for foxes. A p-value of 0.001 indicates this difference is statistically significant.

The AUC results mirror the mean-abundance pattern: rabbits have more total “rabbit-time,” while foxes accumulated substantially less under the crisis scenario.

3.1.3 Extinction Frequency

Condition	Rabbit extinctions	Fisher’s p
Disaster	0 / 5 runs	0.44
Control	2 / 5 runs	

Fisher’s exact test shows this difference is not statistically significant, meaning it could be due to chance.

- **Rabbits.** Mean abundance was 29% higher during the grass-crisis runs ($p < 0.001$). The decline in fox numbers (see below) likely reduced predation pressure, allowing rabbits to flourish even under temporary food scarcity.
- **Foxes.** Mean abundance fell by roughly 50% under the crisis ($p = 0.001$). With fewer rabbits available early in the crisis, many foxes failed to meet their energy requirements.

These results demonstrate that a temporary stop in grass regrowth reshapes community structure: rabbits flourish and foxes decline, yet outright rabbit extinction remains rare in either condition.

4 Conclusion

This project explored the impact of a short-term environmental crisis on predator–prey dynamics using an agent-based simulation. Our findings show that even a temporary disruption in primary resources can lead to a sharp imbalance in trophic structure: prey populations may thrive due to reduced predation, while predator populations suffer from resource shortages. Despite these changes, the system showed resilience, with no consistent trend toward prey extinction.

In the future, several extensions could better this work. Better modeling of grass regrowth rates, seasonality, or agent mobility could improve realism. Adding adaptive behavior, such as predator learning or prey migration, could allow the system to capture more complex survival strategies. Finally, scaling the model to include additional species or multiple interacting habitats could provide insights into broader ecosystem resilience under environmental stress.

References:

- Chen, J., Wang, S. and Liu, Y. (2019) Effects of habitat loss on predator–prey dynamics. *Journal of Theoretical Biology*, 478, pp.1–10. Available at: <https://www.sciencedirect.com/science/article/pii/S0022519319302851>
- Vasconcelos, D., Coelho, H. and Antunes, L. (2020) EcoDynSim: A multi-agent simulation for predator–prey dynamics. *IEEE Access*, 8, pp.163390–163401. Available at: <https://ieeexplore.ieee.org/document/9206765>

- Venturino, E. (2014) How environmental changes affect predator–prey dynamics. Bulletin of Mathematical Biology, 76(7), pp.1649–1674. Available at: <https://link.springer.com/article/10.1007/s11538-014-0060-6>
- Wilensky, U. and Rand, W. (2018) An introduction to agent-based modeling: Modeling natural, social, and engineered complex systems with NetLogo. MIT Press. Available at: <https://arxiv.org/abs/1805.11816>